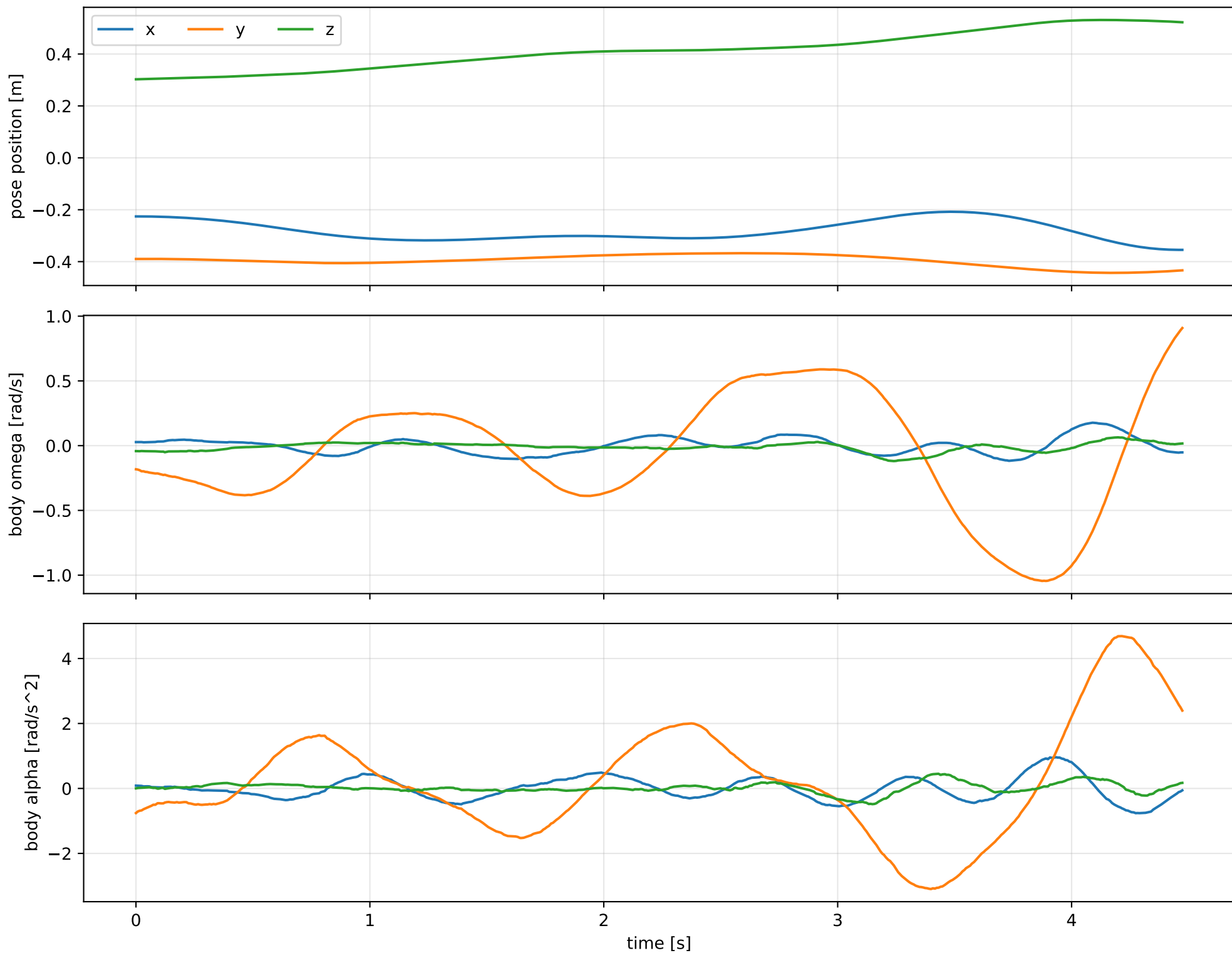
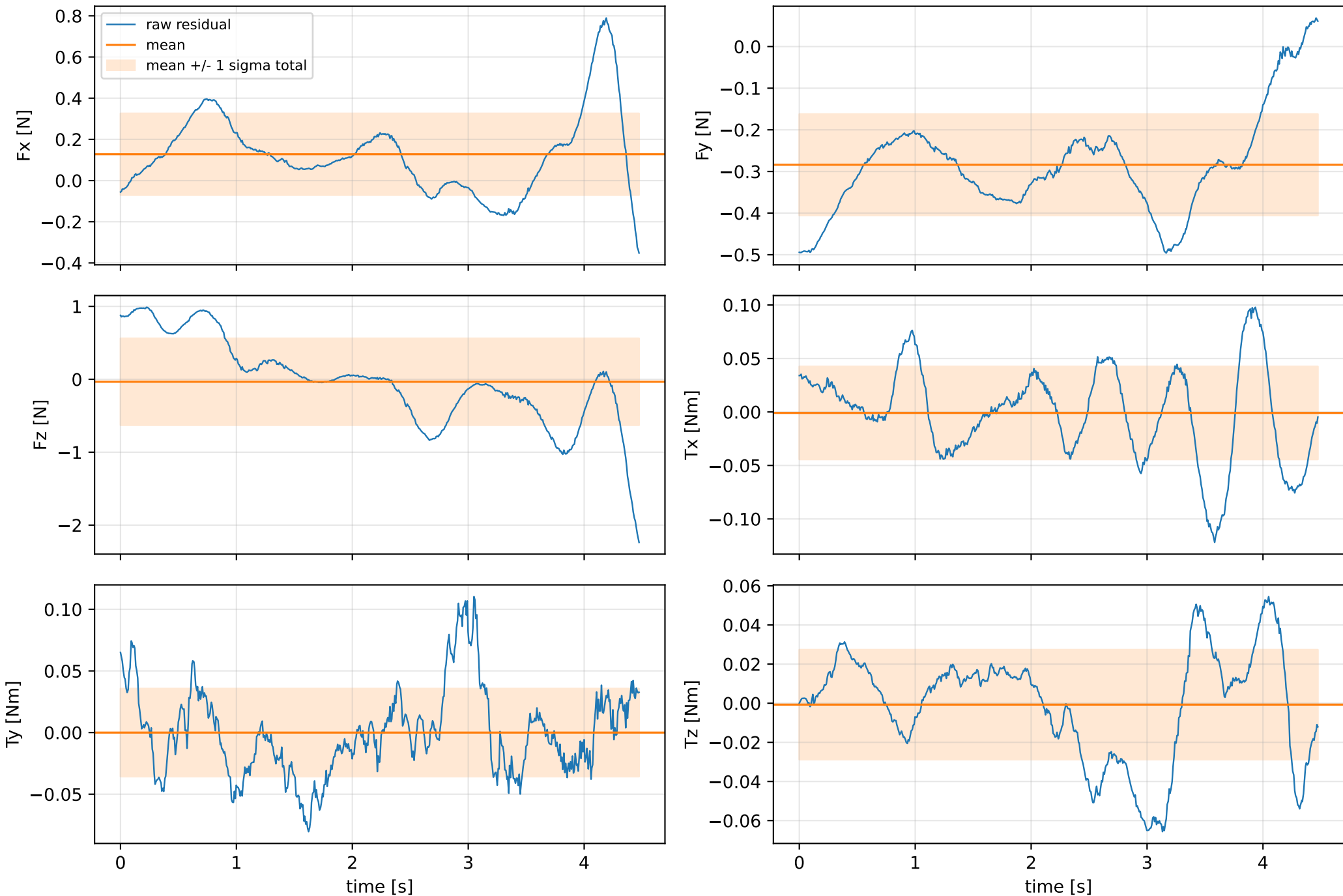


inertia\_over\_mass\_zz\_nominal: SG trajectory and derived motion

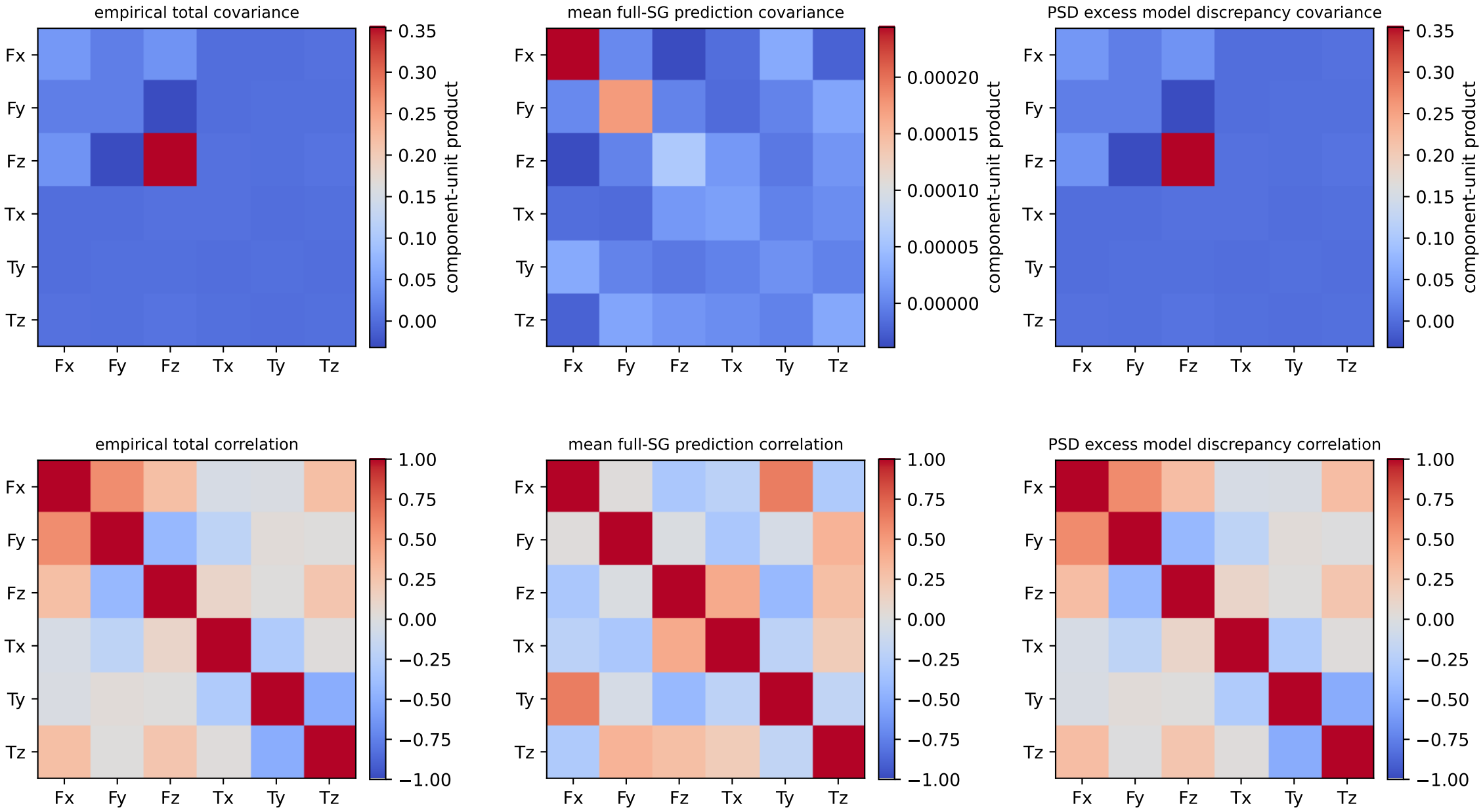


inertia\_over\_mass\_zz\_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)  
mass gauge fixed to nominal mass = 2.35159759 kg



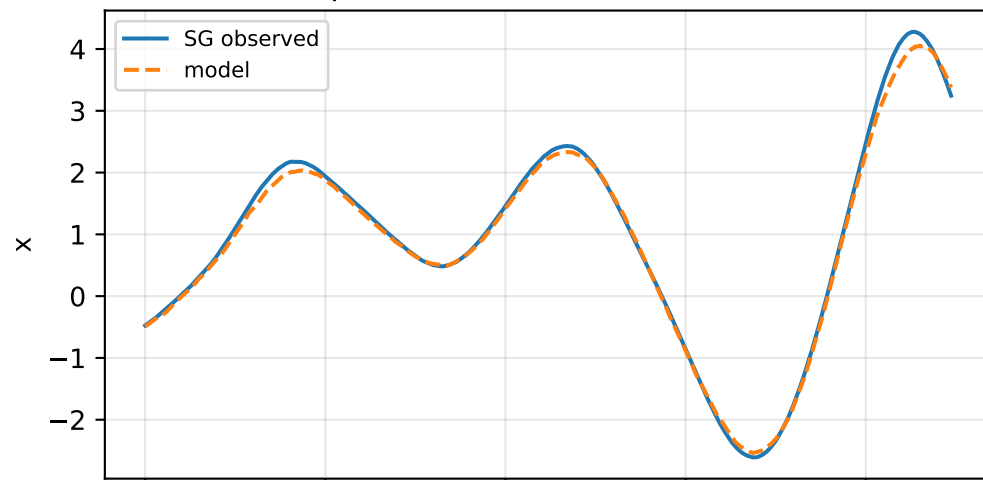
vector RMS about zero: force 0.71105 N, torque 0.062773 Nm; component std about mean: force [0.1982 0.1217 0.5951], torque [0.0434 0.0357 0.0281]

inertia\_over\_mass\_zz\_nominal: residual-wrench covariance decomposition in nominal-mass gauge  
force [N], torque [Nm]; raw excess eigenvalues [3.455e-04 1.282e-03 1.923e-03 3.508e-03 4.420e-02 3.604e-01]

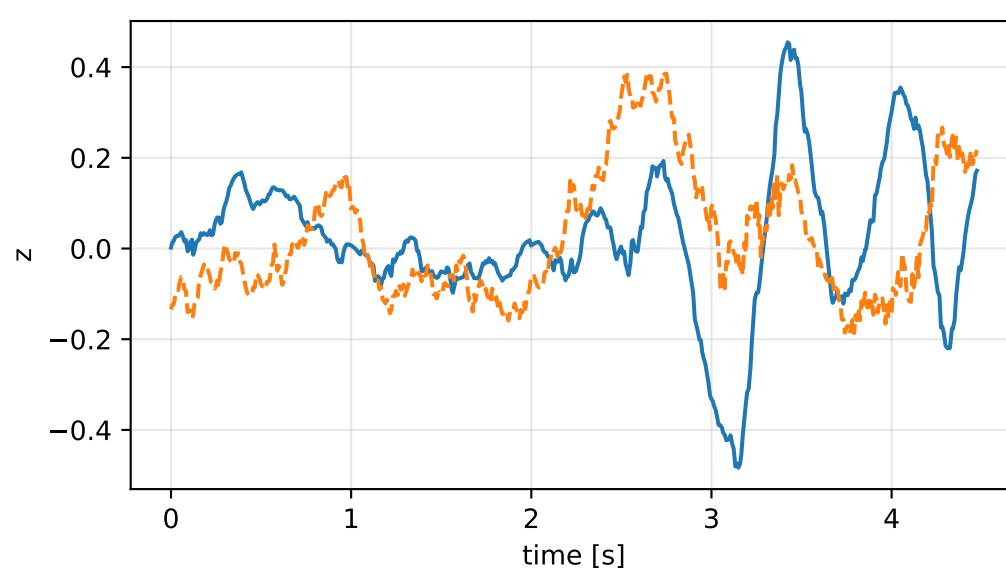
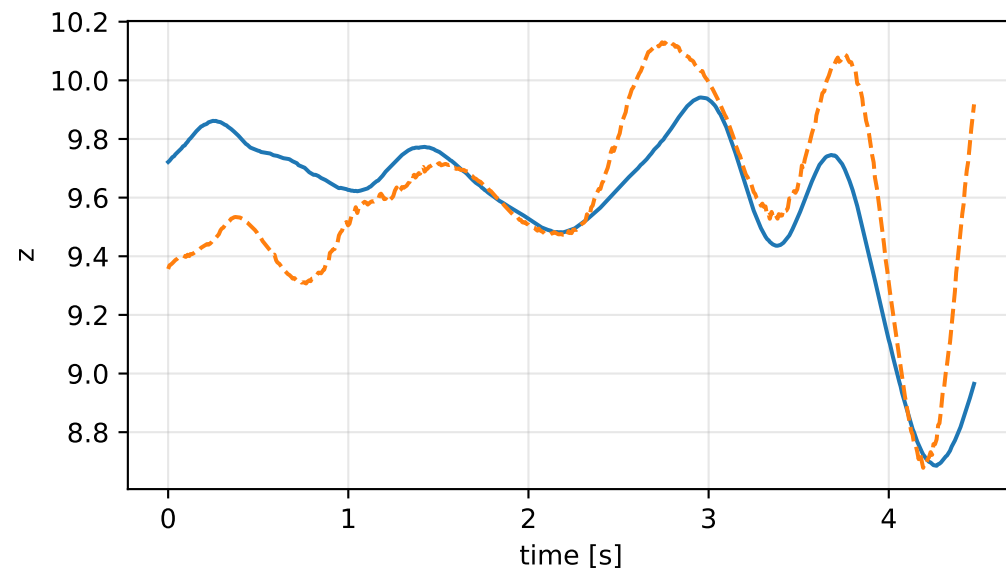
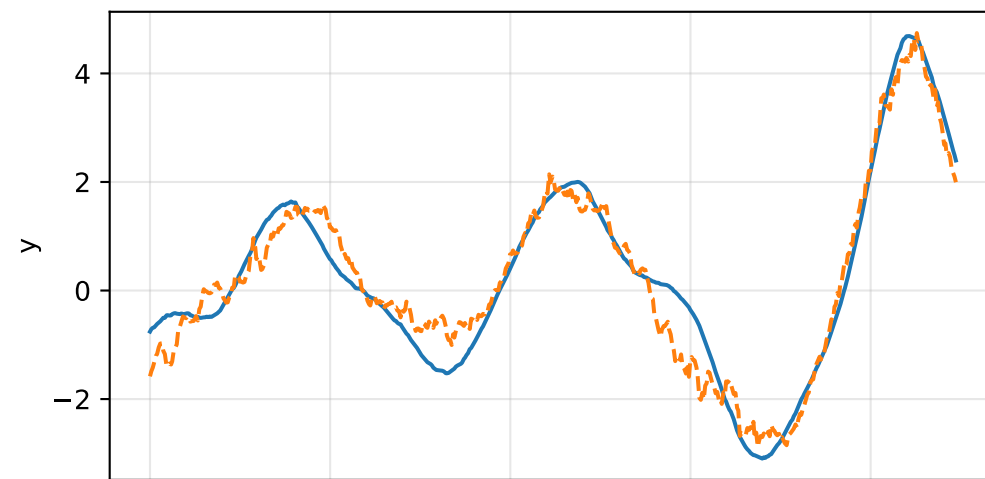
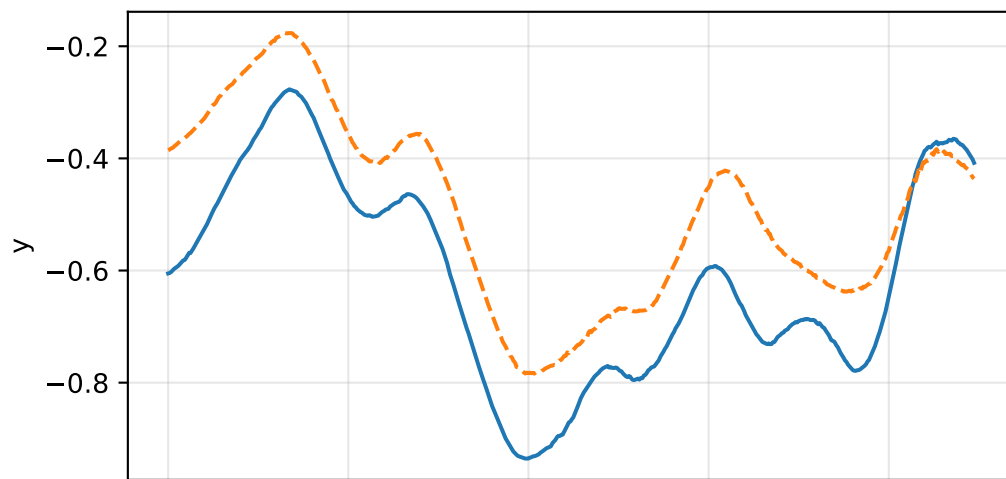
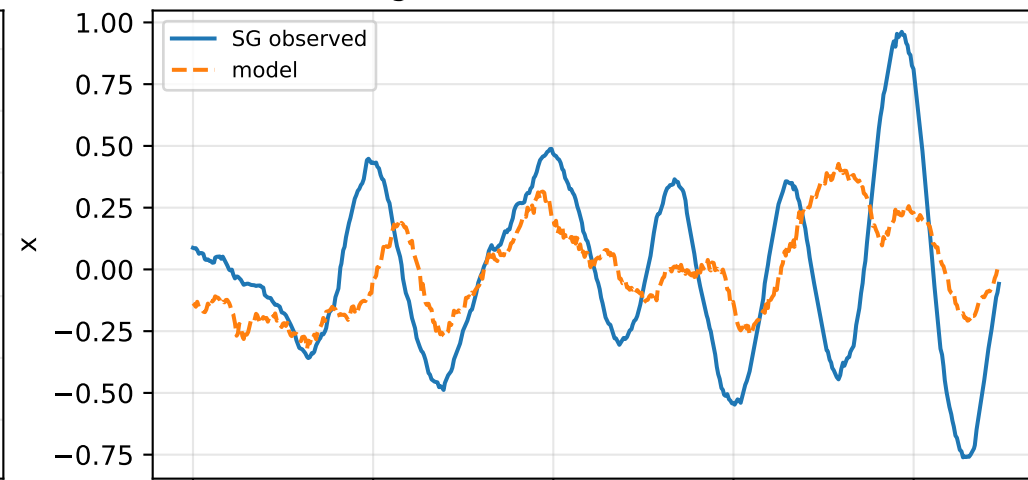


# inertia\_over\_mass\_zz\_nominal: acceleration residual objective

## specific acceleration [m/s^2]

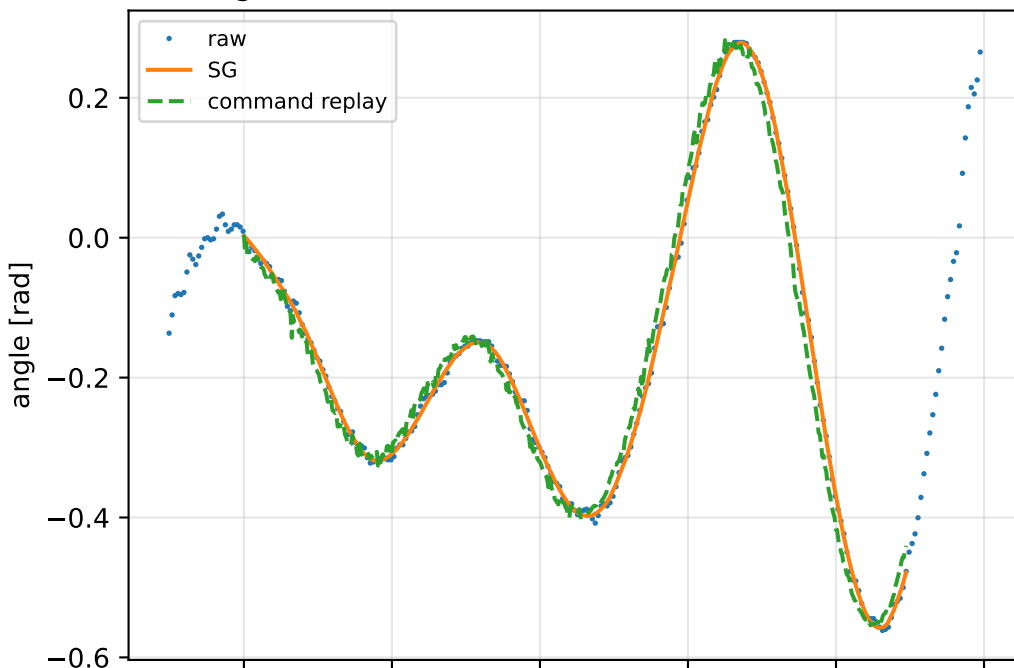


## angular acceleration [rad/s^2]

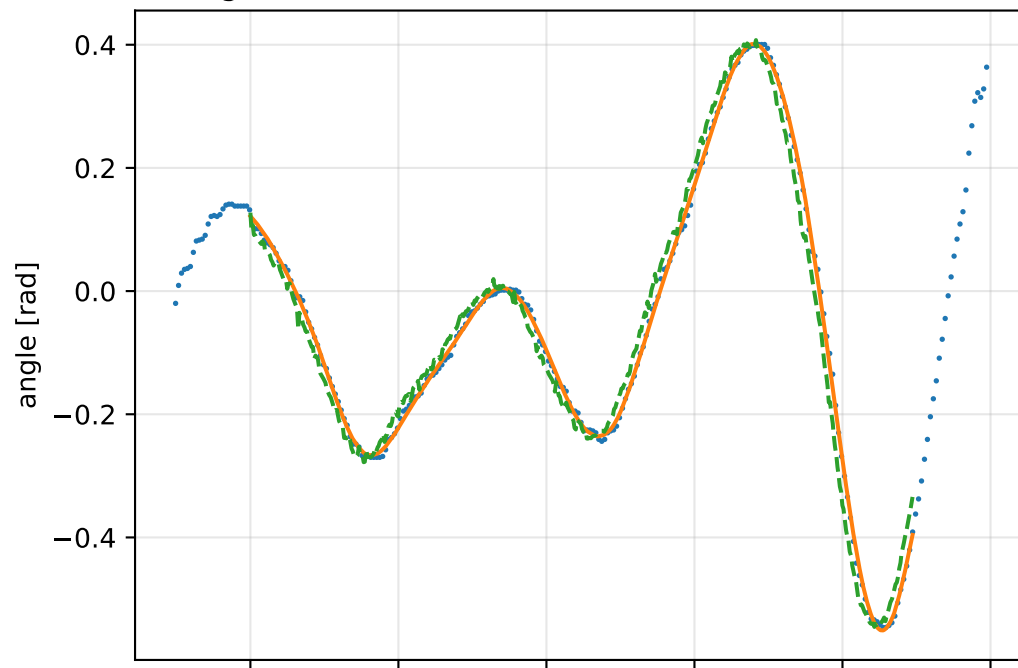


inertia\_over\_mass\_zz\_nominal: gimbal measurement audit (objective=measured\_sg)

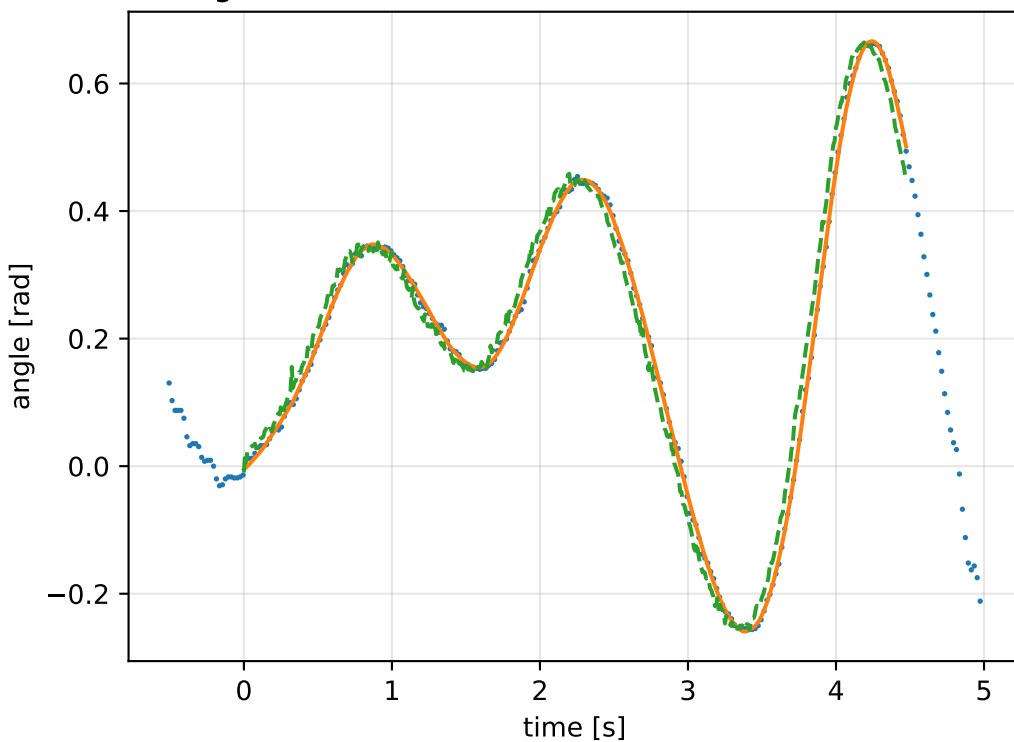
gimbal 1: RMSE=0.027, max=0.07529 rad



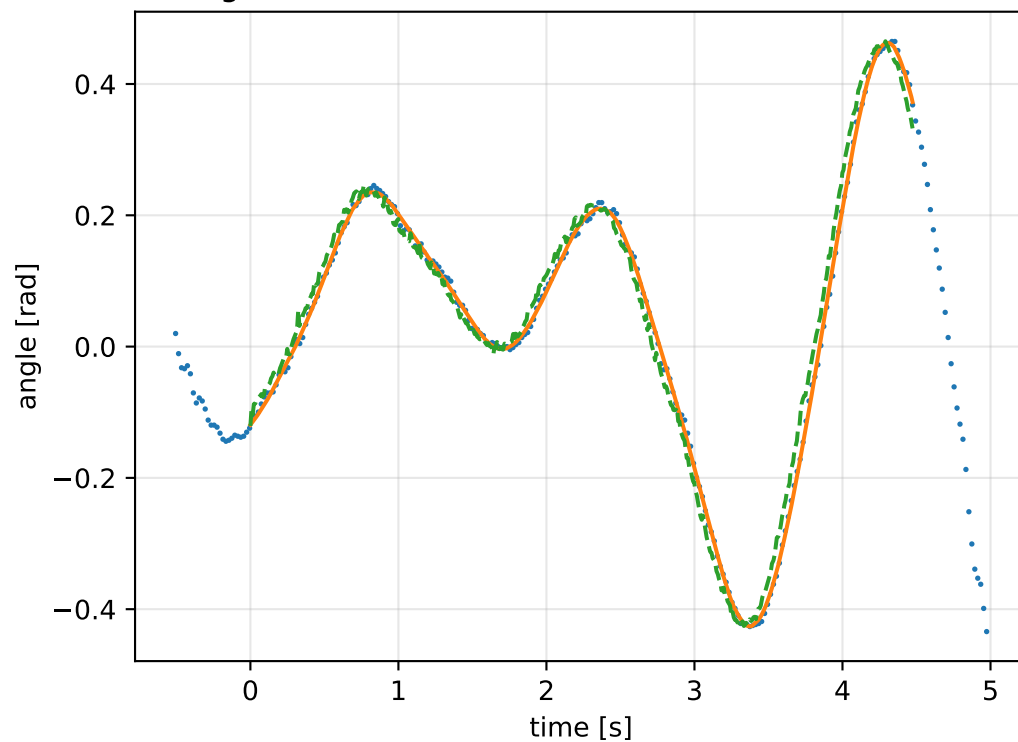
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

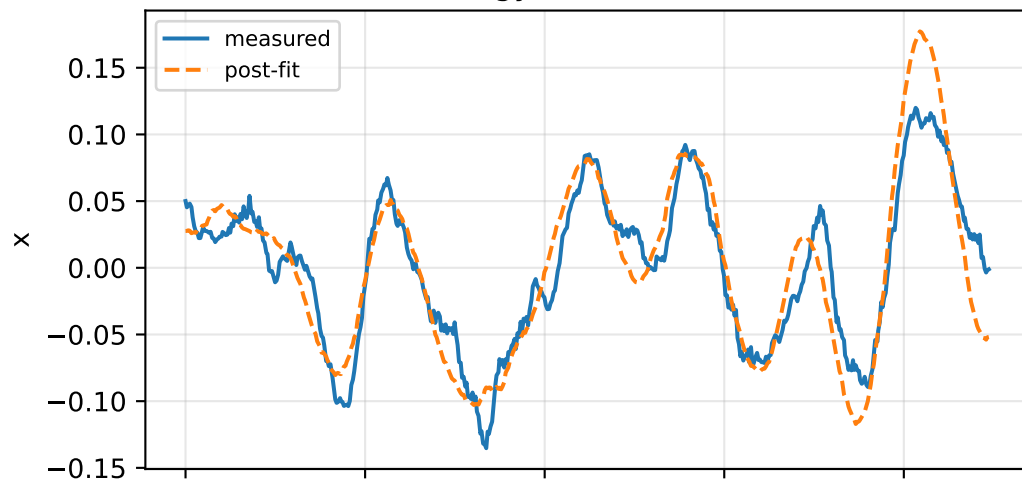


gimbal 4: RMSE=0.02598, max=0.07261 rad

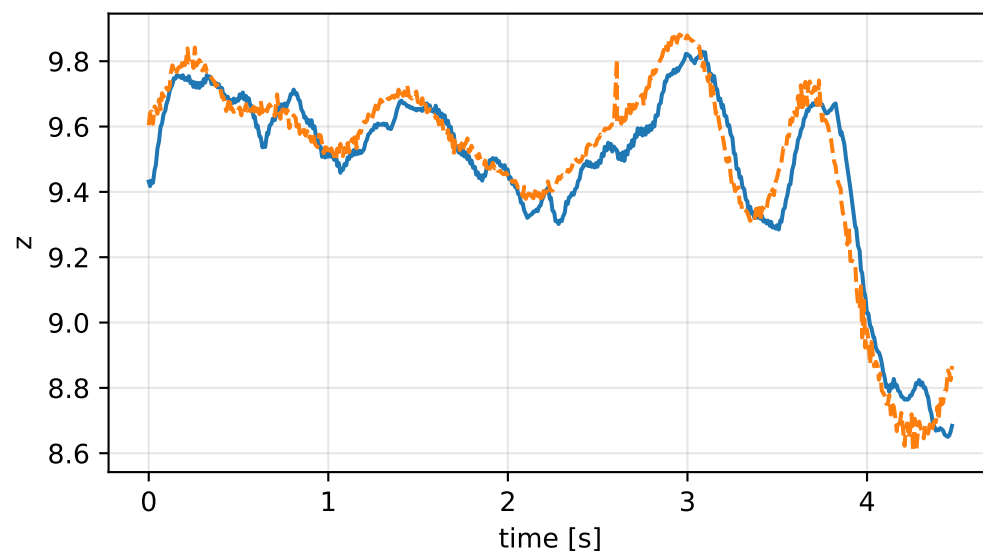
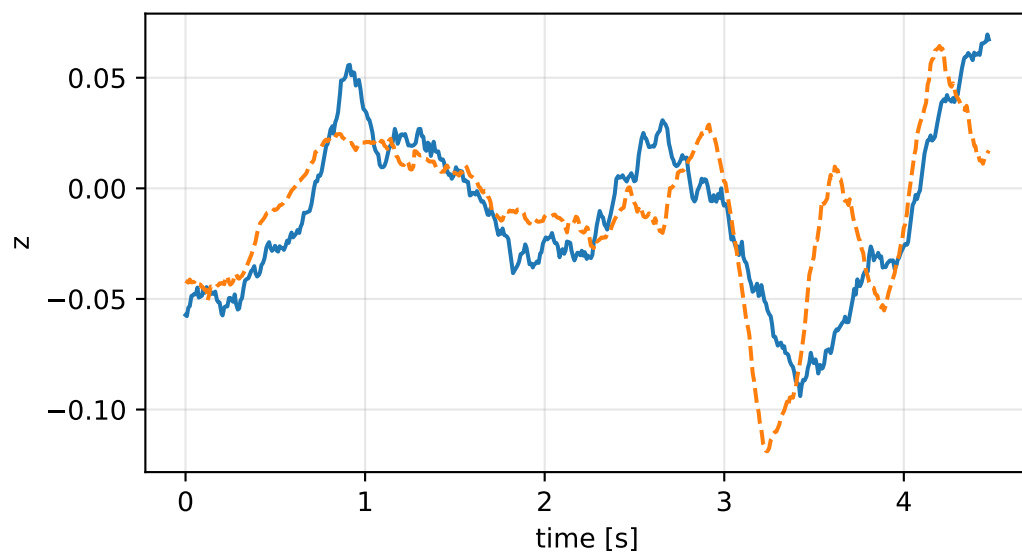
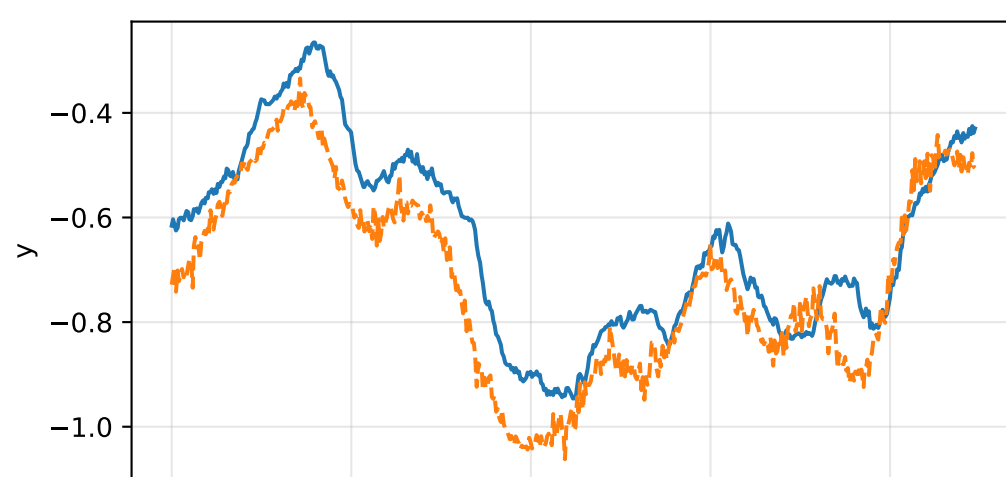
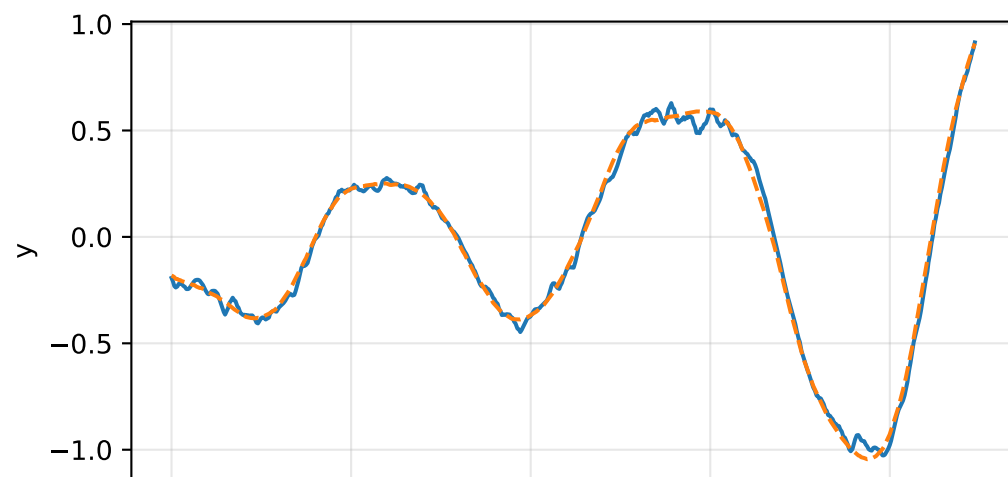
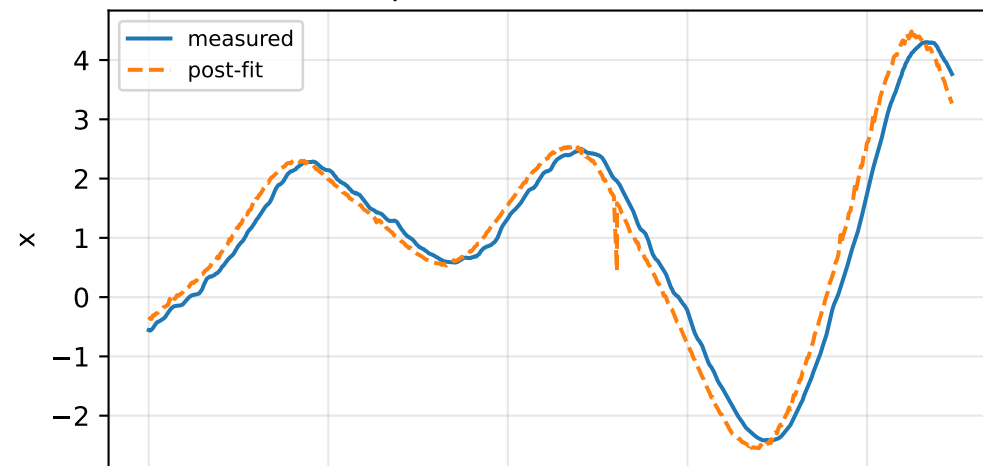


# inertia\_over\_mass\_zz\_nominal: IMU comparison (diagnostic only)

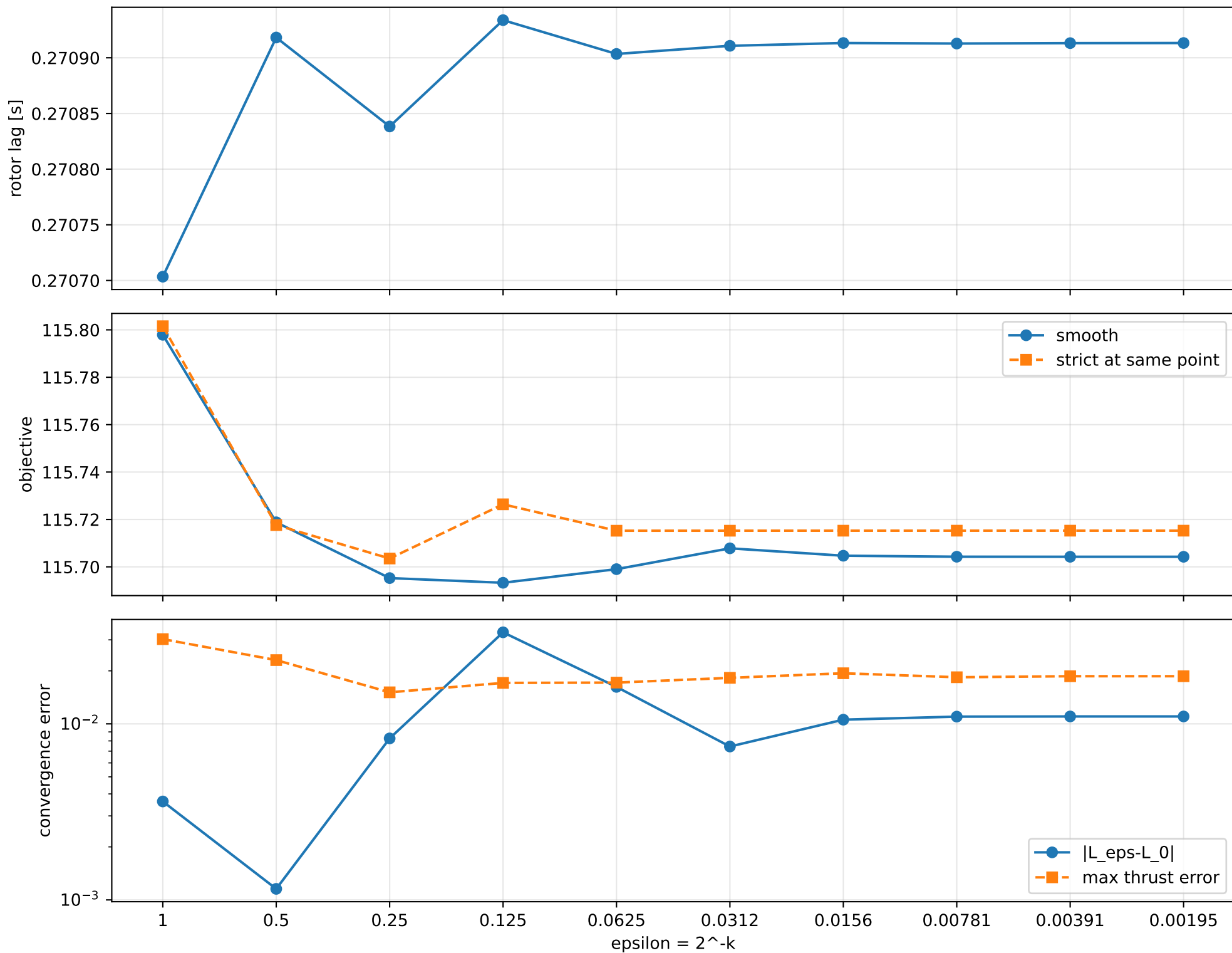
## gyro [rad/s]



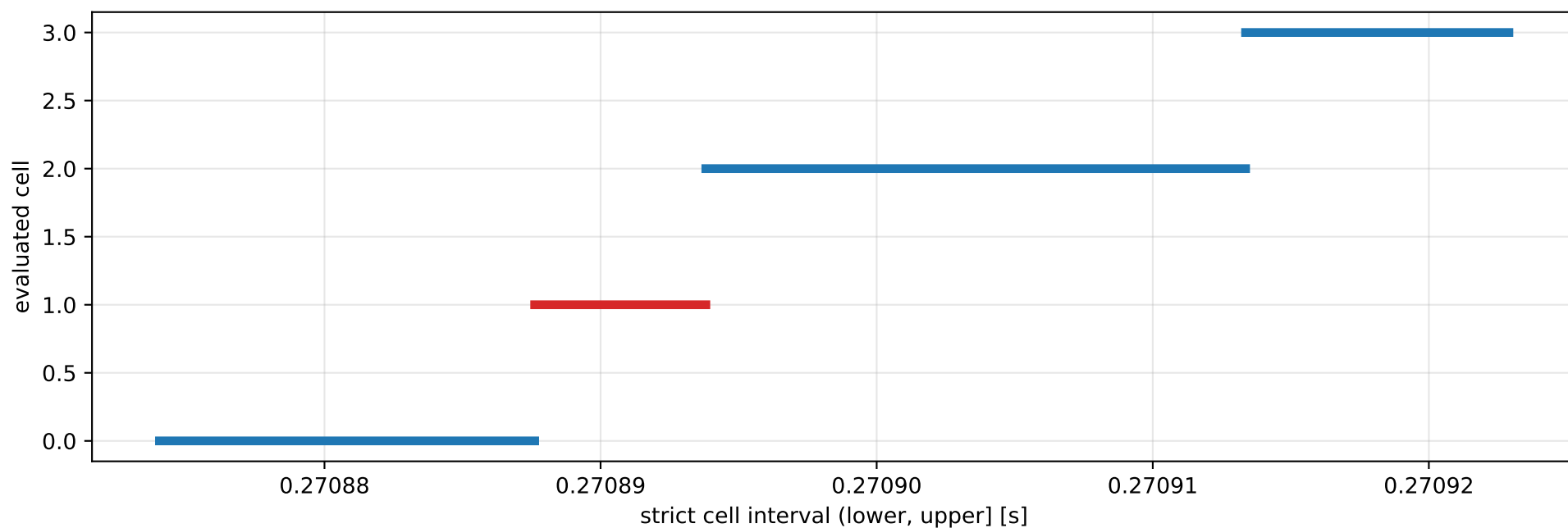
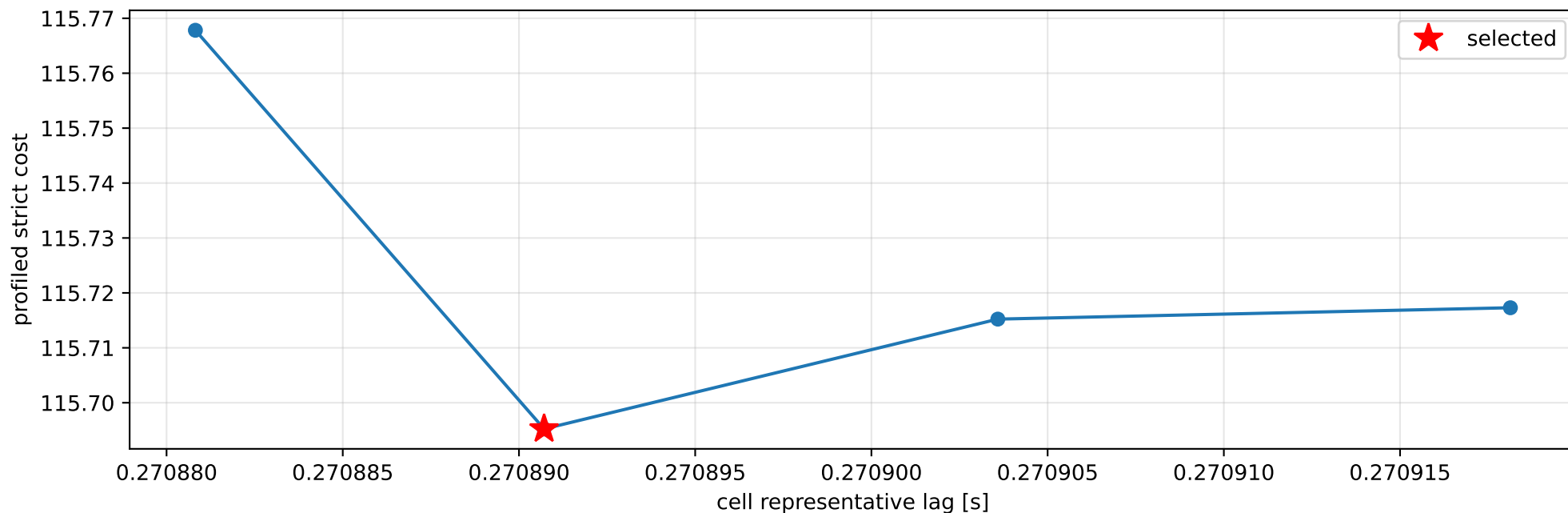
## specific force [m/s^2]



inertia\_over\_mass\_zz\_nominal: smooth-to-strict continuation



inertia\_over\_mass\_zz\_nominal: exact strict-ZOH lag cells



selected (0.270887613, 0.270893812] s; final smooth 0.270913251 s; data boundary=False



## inertia\_over\_mass\_zz\_nominal: parameter estimate

Case: inertia\_over\_mass\_zz\_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.54277738

inertia [kg m<sup>2</sup>]:

```
[[ 0.09135216  0.00372005 -0.00932515]
 [ 0.00372005  0.0526358  -0.01144238]
 [-0.00932515 -0.01144238  0.08462592]]
```

CoG body [m]: [-0.02427499 0.00924108 0.01008838]

force effectiveness: [0.58192248 0.49460415 0.51894142 0.62841705]

scale-free J/m [m<sup>2</sup>]:

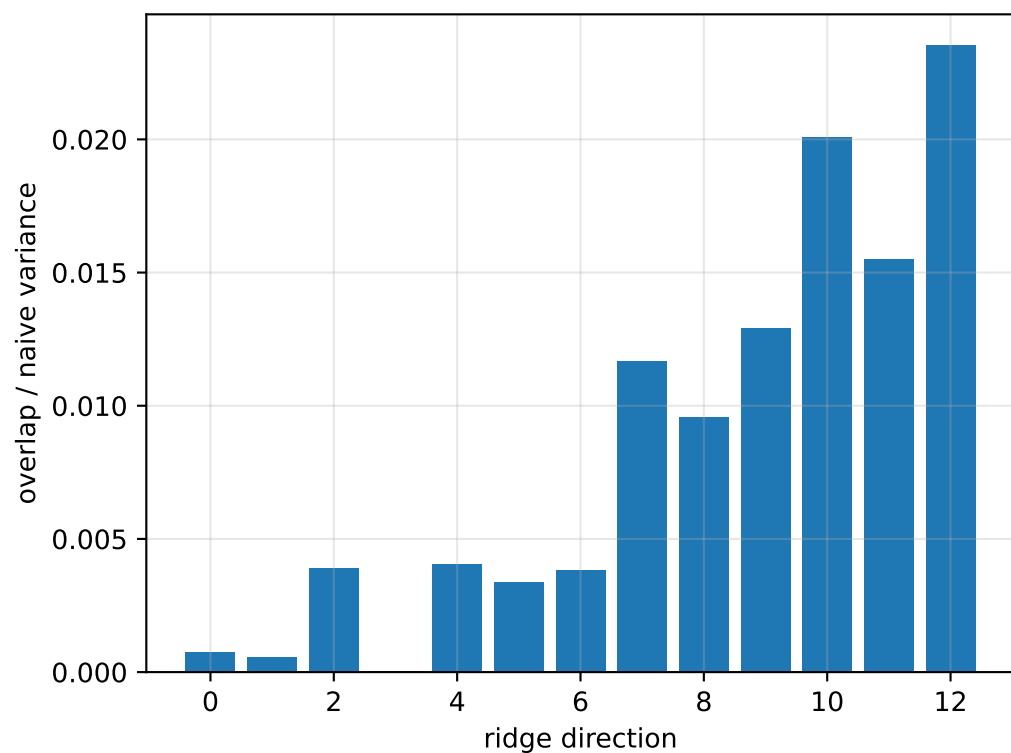
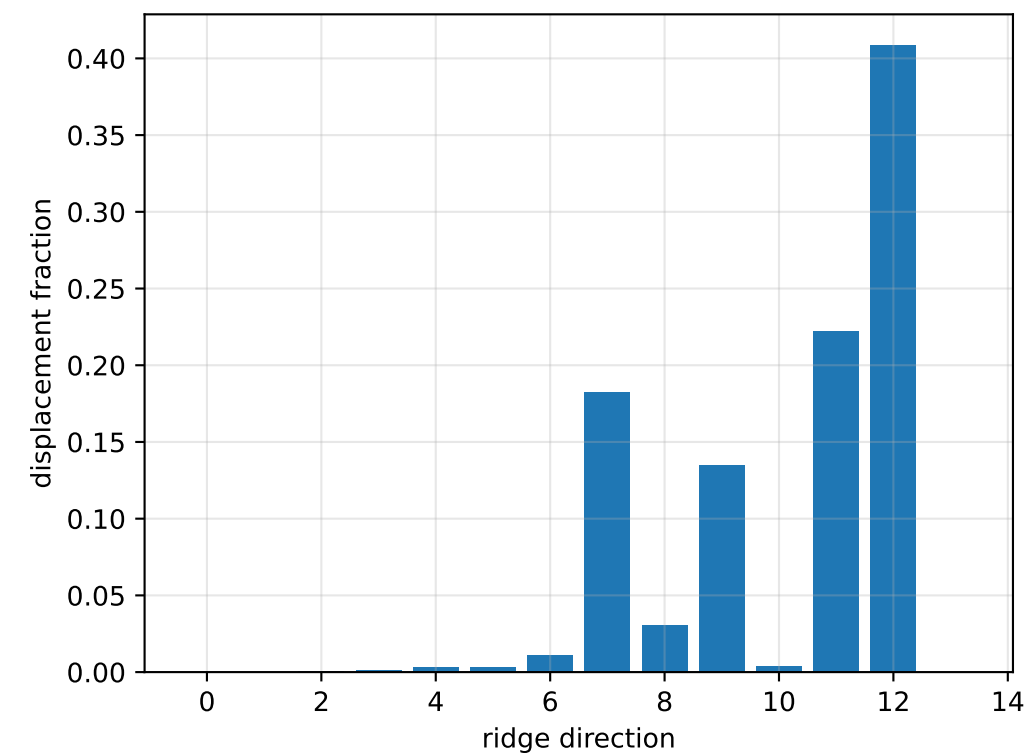
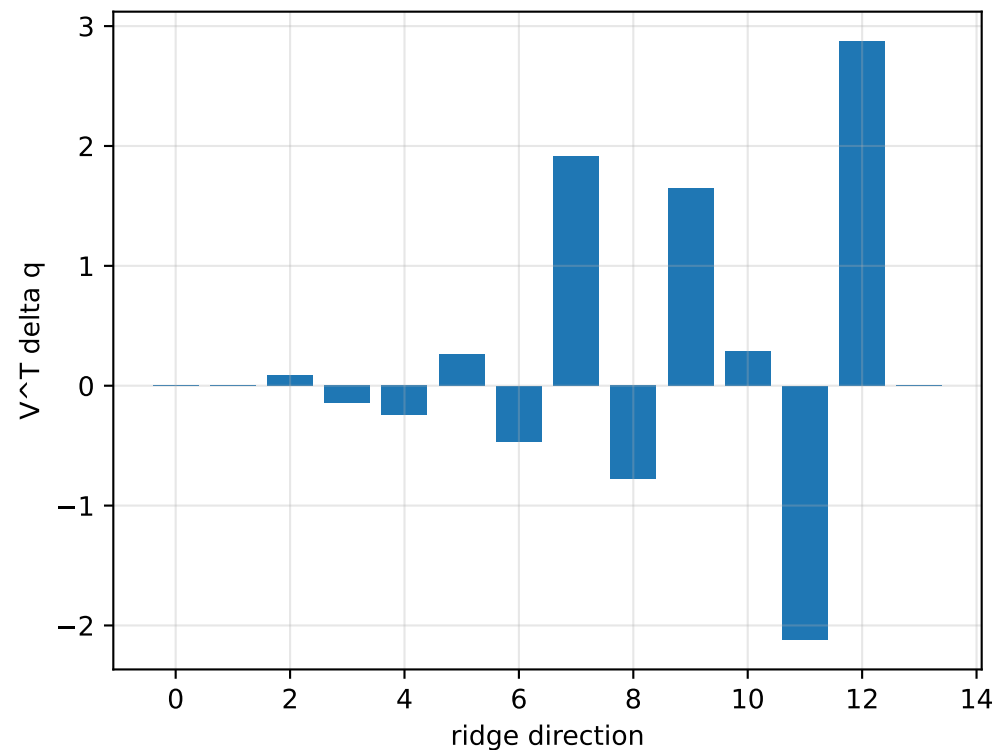
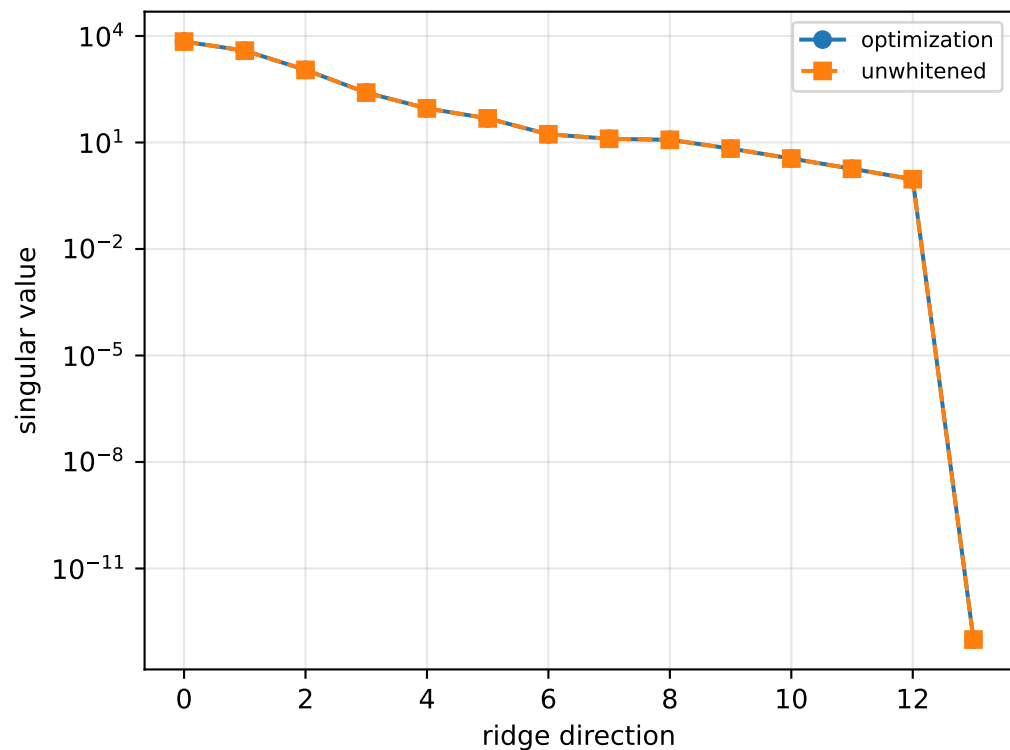
```
[[ 0.05921279  0.00241127 -0.00604439]
 [ 0.00241127  0.03411756 -0.00741674]
 [-0.00604439 -0.00741674  0.05485297]]
```

scale-free f/m: [0.37719148 0.32059334 0.33636831 0.40732841]

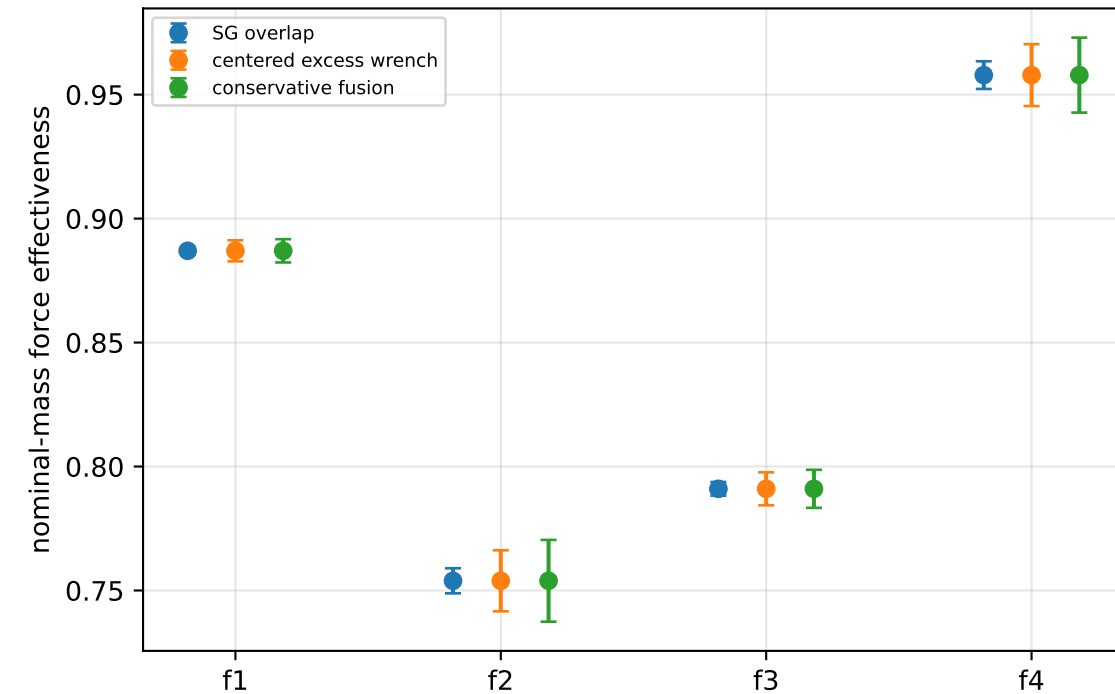
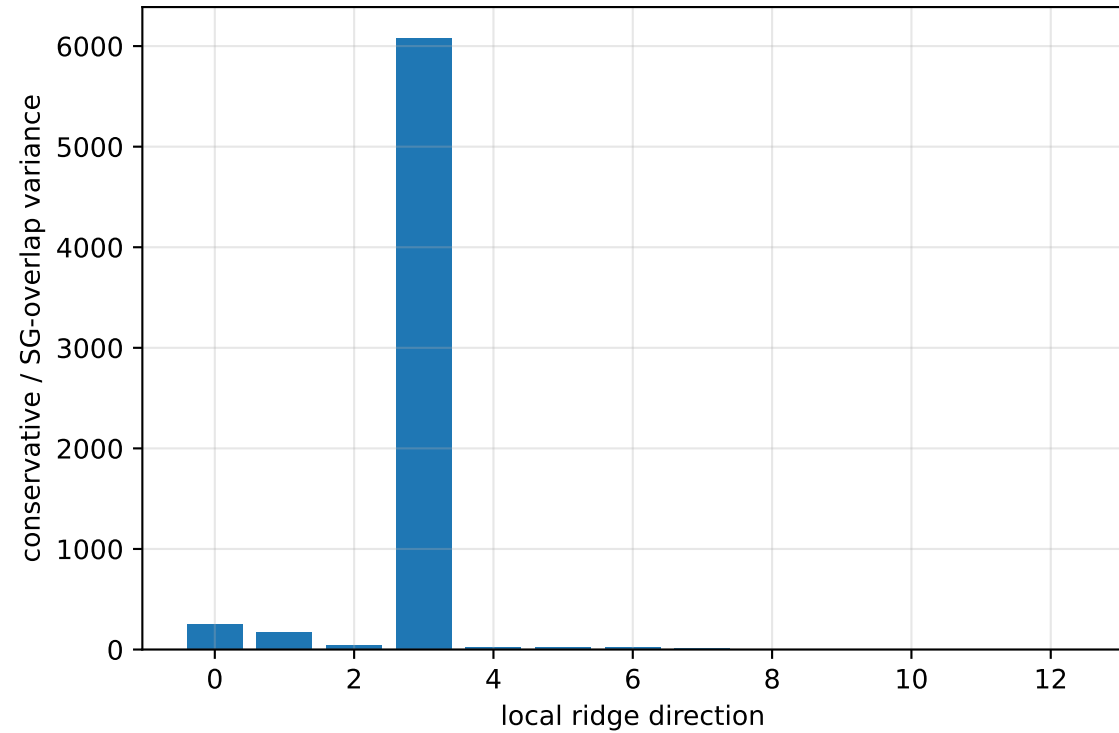
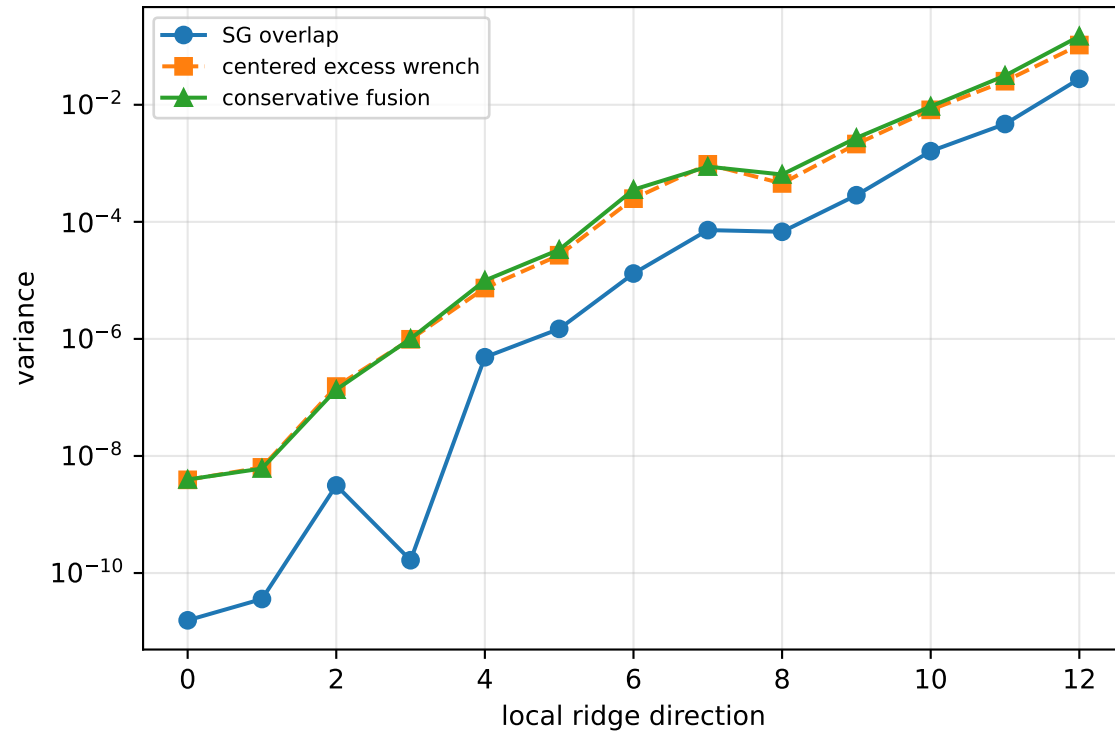
rotor lag cell: (0.270887613, 0.270893812] s

representative: 0.270890713 s

inertia\_over\_mass\_zz\_nominal: ridge spectrum (rank 13, nullity 1, || $\mathbf{v}$ ||=1.47e-13)



# inertia\_over\_mass\_zz\_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions  
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 219.3$   
wrench-to-acceleration recovery max error =  $2.83\text{e-}15$   
exact scale gauge ridge index = 13 (alignment 1)

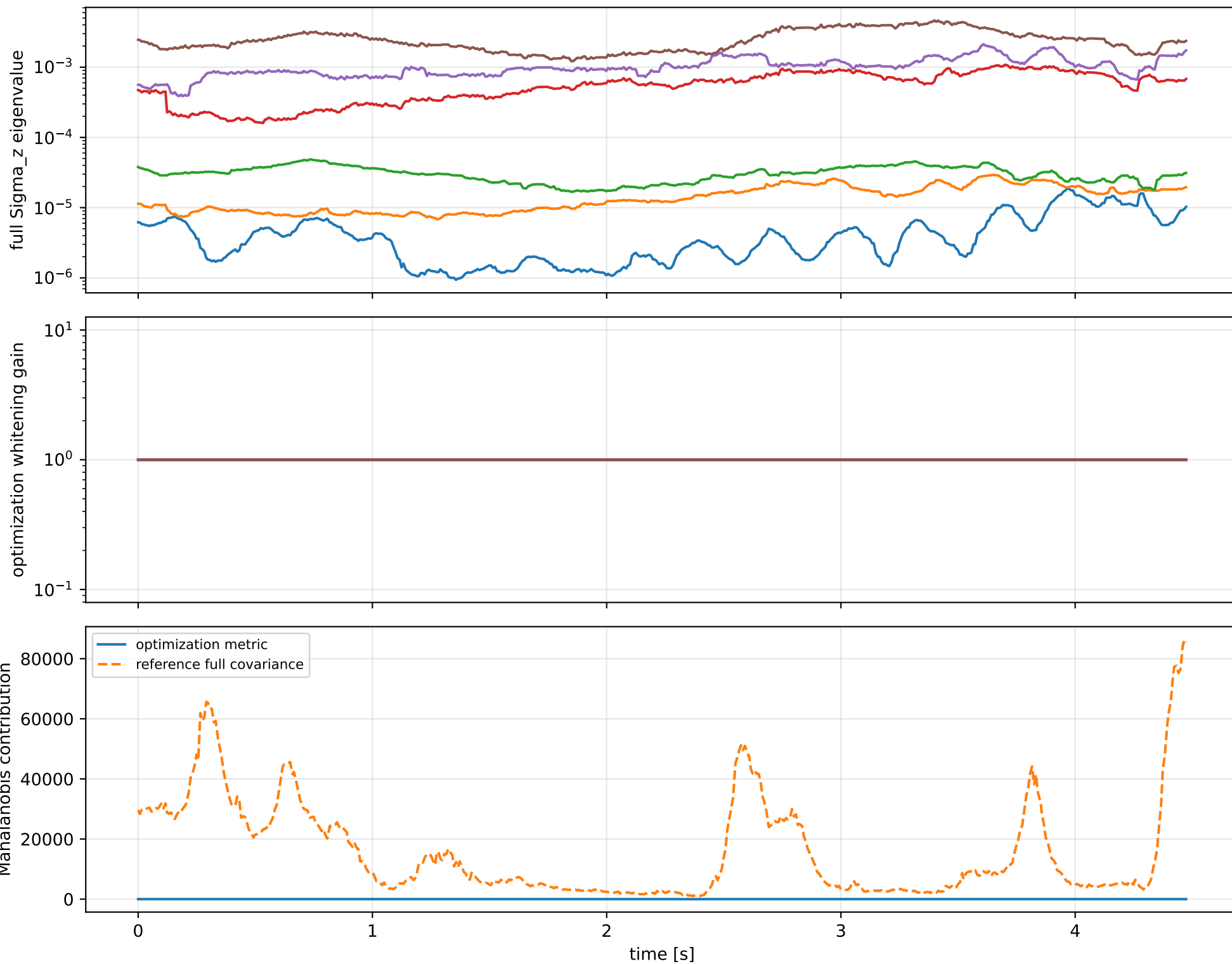
index 12: variance=0.1481, ratio=5.339  
 $\log_{\text{mass}}=+0.061$ ; second\_moment\_diag\_1=-0.824  
second\_moment\_diag\_2=+0.0372; second\_moment\_diag\_3=+0.481  
second\_moment\_offdiag\_12=-0.199; second\_moment\_offdiag\_13=+0.155  
second\_moment\_offdiag\_23=-0.0667; cog\_x=+0.000362  
cog\_y=-0.000341; cog\_z=+1.3e-05  
log\_force\_eff\_1=+0.0589; log\_force\_eff\_2=+0.0672  
log\_force\_eff\_3=+0.0587; log\_force\_eff\_4=+0.0601

index 11: variance=0.03147, ratio=6.711  
 $\log_{\text{mass}}=+0.1$ ; second\_moment\_diag\_1=-0.0405  
second\_moment\_diag\_2=+0.057; second\_moment\_diag\_3=-0.527  
second\_moment\_offdiag\_12=-0.459; second\_moment\_offdiag\_13=+0.663  
second\_moment\_offdiag\_23=+0.116; cog\_x=+0.00374  
cog\_y=-0.00182; cog\_z=+0.00154  
log\_force\_eff\_1=+0.0966; log\_force\_eff\_2=+0.123  
log\_force\_eff\_3=+0.114; log\_force\_eff\_4=+0.0759

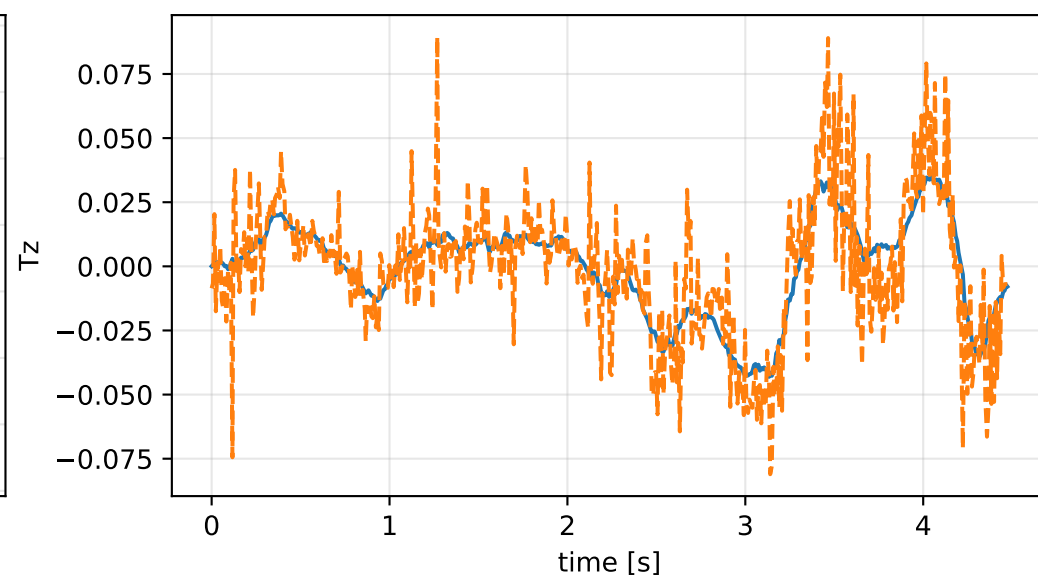
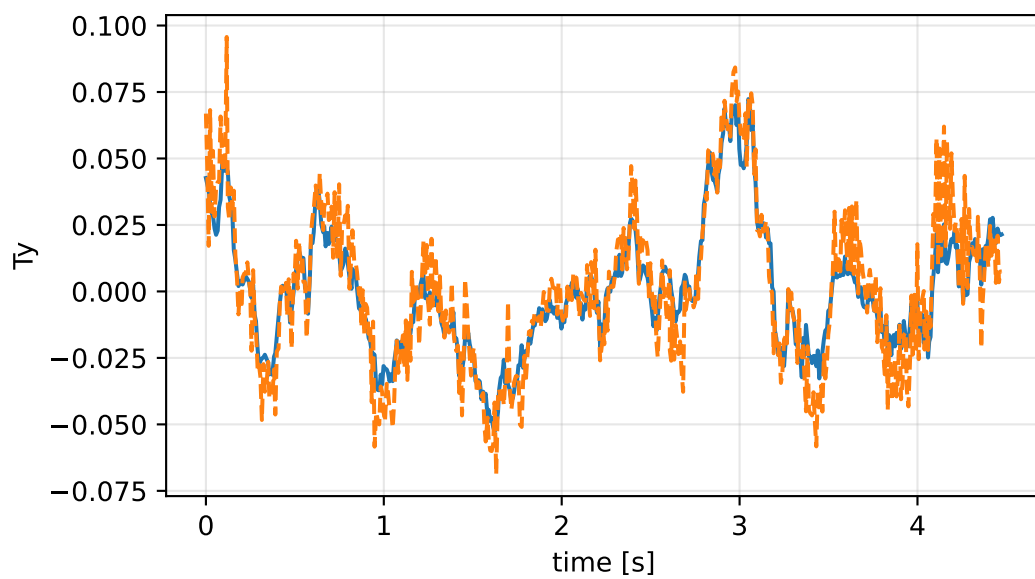
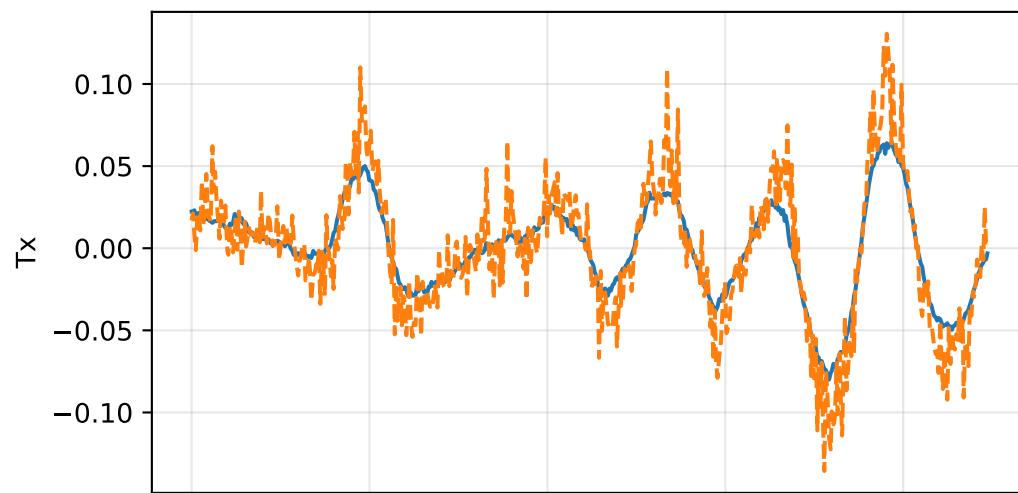
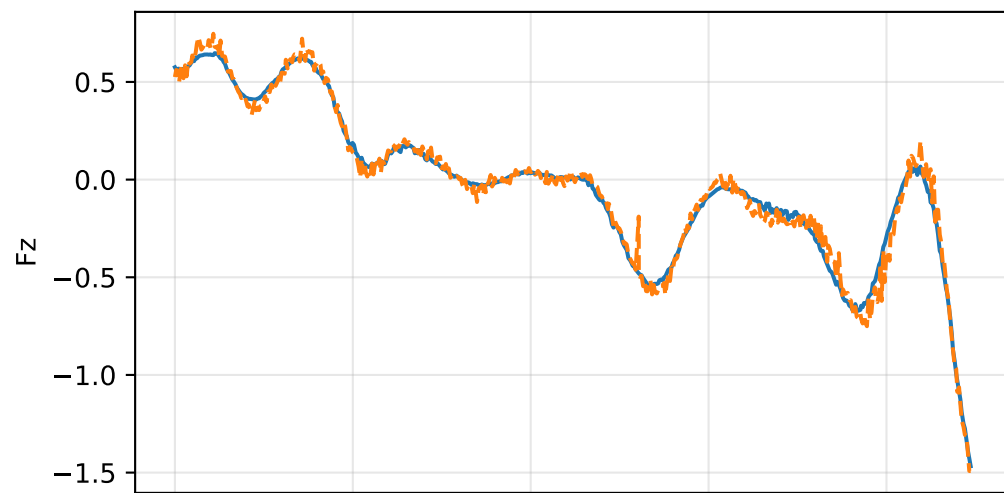
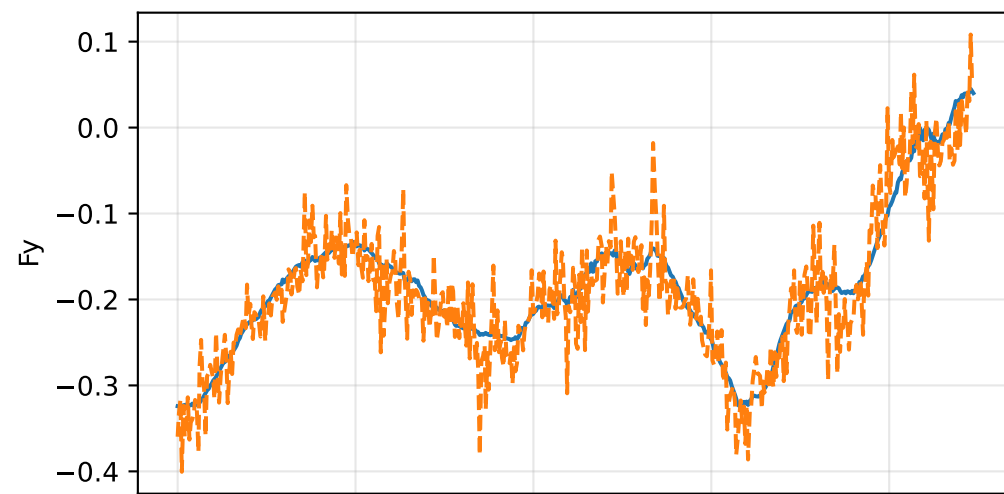
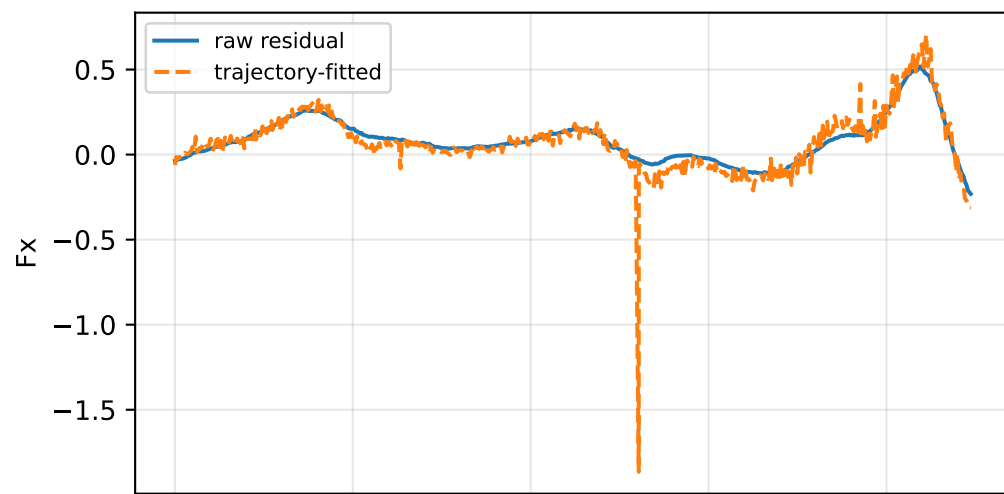
index 10: variance=0.009369, ratio=5.814  
 $\log_{\text{mass}}=+0.125$ ; second\_moment\_diag\_1=+0.133  
second\_moment\_diag\_2=-0.925; second\_moment\_diag\_3=+0.156  
second\_moment\_offdiag\_12=-0.137; second\_moment\_offdiag\_13=+0.0172  
second\_moment\_offdiag\_23=-0.00207; cog\_x=+0.00464  
cog\_y=-0.00134; cog\_z=-0.000871  
log\_force\_eff\_1=+0.111; log\_force\_eff\_2=+0.157  
log\_force\_eff\_3=+0.141; log\_force\_eff\_4=+0.102

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia\_over\_mass\_zz\_nominal: optimization vs reference covariance



inertia\_over\_mass\_zz\_nominal: wrench / closure (force max 4.22e-15, torque max 5.38e-17)



## inertia\_over\_mass\_zz\_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo\_condition\_inertia\_over\_mass\_zz\_nominal

role: pseudo\_conditioning\_ablation

source: /home/leus/catkin\_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo\_conditioning/scalar/inertia\_over\_ma

SHA256: 0aabc66d65b6ca544bf367511021d32b9f332e43a03fe96c3ce28f09a6e7e577

data objective: 115.695233629

prior objective: 1.37164190517e-08

total objective: 115.695233643

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia\_over\_mass\_zz\_nominal (inertia\_over\_mass\_m2)

components: zz

target: [0.05485297]

std: [1.e-06]

final: [0.05485297]

error: [1.65628615e-10]

standardized residual: [0.00016563]

factor objective: 1.37164190517e-08